



NOTES OF CLASS 10TH CHEMISTRY

Chp#4

Electrochemistry

SECTION – A

Electrochemical cells

1. In a Galvanic cell, which of the following statements is true?

- a. Electrons flow from cathode to anode b. Oxidation occurs at the cathode
c. The anode is negative d. Reduction occurs at the anode

2. What occurs at the cathode of a Galvanic cell?

- a. Oxidation **b. Reduction** c. Electron loss d. None of the above

3. Which of the following is used to maintain electrical neutrality in an electrochemical cell?

- a. Electrode b. Wire **c. Salt bridge** d. Battery

4. In an electrolytic cell, the direction of electron flow is:

- a. From cathode to anode **b. From anode to cathode** c. Alternates periodically d. Random

5. Which one of these is a correct example of a Galvanic cell?

- a. Copper plating on iron b. Electrolysis of water **c. Daniel cell** d. Battery charging

6. Which of following is incorrect for spontaneous reaction?

- a. self-occurring **b. requires energy** c. produces energy d. All

7. Which of the following is spontaneous cell?

- a. Electrolytic cell **b. galvanic cell** c. down's cell d. all

8. A cell that converts electrical energy into chemical energy is called:

- a. Electrolytic cell** b. Galvanic cell c. Daniel cell d. all of these

9. In an electrolytic cell, the flow of electrons is:

- a. From cathode to anode externally **b. From anode to cathode externally**
c. Random d. There is no flow of electrons

10. In an electrolytic cell, which of the following occurs at the cathode?

- a. Oxidation b. Electron loss **c. Reduction** d. Gas evolution only

11. The energy required to operate an electrolytic cell comes from:

- a. Light b. A galvanic cell **c. External electric power source** d. Heat

12. In the electrolysis of molten PbCl_2 , which ion is reduced at the cathode?

- a. Cl^- **b. Pb^{2+}** c. O^{2-} d. H^+

13. What is observed at the cathode during the electrolysis of molten PbCl_2 ?

- a. Bubbles of gas b. Chlorine gas **c. Molten lead metal** d. Oxygen gas

14. Which gas is released at the anode?

- a. Oxygen **b. Chlorine** c. Hydrogen d. Nitrogen

15. Which of the following best describes the role of graphite electrodes in this electrolysis?

- a. They take part in the chemical reaction b. They are oxidized
c. They are inert and only conduct electrons d. They dissolve in molten PbCl_2

16. In the electrolysis of a dilute aqueous solution of NaCl , what is produced at the cathode?

- a) Sodium metal **b) Hydrogen gas** c) Chlorine gas d) Oxygen gas

17. What is the product at the anode during the electrolysis of a dilute NaCl solution?

- a) Chlorine gas **b) Oxygen gas** c) Sodium metal d) Hydrogen gas

18. Which ions are preferentially discharged at the cathode in an aqueous solution?

- a) Na^+ **b) H^+** c) Cl^- d) OH^-

19. Why is hydrogen gas liberated at the cathode instead of sodium in dilute NaCl solution?

- a) Sodium is more reactive than hydrogen **b) Water undergoes preferential reduction**
c) Hydrogen ions are present in greater concentration d) All of the above

20. During the electrolysis of dilute sulfuric acid, what is formed at the cathode?

- a) Hydrogen gas** b) Oxygen gas c) Sulfur d) Water

21. What is formed at the anode during the electrolysis of dilute sulfuric acid?

- a) Sulfur dioxide b) Hydrogen gas **c) Oxygen gas** d) Sulfuric acid

22. Which ions are responsible for the formation of oxygen at the anode?

- a) H^+ **b) OH^-** c) SO_4^{2-} d) O^{2-}

23. What is observed at the cathode during electrolysis of CuSO_4 using copper electrodes?

- a) Formation of hydrogen gas **b) Formation of copper metal**
c) Release of oxygen gas d) No reaction

24. The mass of the cathode will:

- a) Increase** b) Decrease c) Stay the same d) First increase,
then decrease

25. The mass of the anode will:

- a) Increase **b) Decrease** c) Stay the same d) Double



Electrolytes

26. What is an electrolyte?

- a. A substance that burns in air
- b. A substance that dissolves in water to give a non-conducting solution
- c. A substance that dissolves in water to conduct electricity
- d. A substance that forms precipitate in water

27. Which of the following is a strong electrolyte?

- a. Glucose
- b. Acetic acid
- c. Sodium chloride
- d. Ammonia

28. Which of the following is a non-electrolyte?

- a. KCl
- b. NaOH
- c. Sugar ($C_6H_{12}O_6$)
- d. HCl

29. In aqueous solution, strong electrolytes are:

- a. Partially ionized
- b. Completely ionized
- c. Not ionized
- d. Ionized only in hot water

30. Which of the following is a weak electrolyte?

- a. H_2SO_4
- b. HCl
- c. CH_3COOH
- d. NaCl

31. The degree of ionization of an electrolyte increases with:

- a. Decrease in temperature
- b. Increase in concentration
- c. Dilution
- d. Pressure

32. What type of electrolyte is molten $PbCl_2$?

- a. Weak electrolyte
- b. Strong electrolyte
- c. Non-electrolyte
- d. Weak acid

ELECTROPLATING

33. Zinc plating on steel is done by using Zinc metal as.

- a. Cathode
- b. Anode
- c. Both
- d. None

34. The electrolyte used during zinc plating is:

- a. $K_2[Zn(CN)_4]$
- b. H_2SO_4
- c. $SnSO_4$
- d. None

35. During zinc plating which of the following prevents the hydrolysis of electrolyte.

- a. $K_2[Zn(CN)_4]$ b. H_2SO_4 **c. NaCN** d. None

36. . Tin plating on steel is done by using Tin metal.

- a. Cathode **b. Anode** c. Both d. None

37. .The electrolyte used during Tin plating is:

- a. $K_2[Zn(CN)_4]$ b. H_2SO_4 **c. $SnSO_4$** d. None

38. During Tin plating which of the following prevents the hydrolysis of electrolyte.

- a. $K_2[Zn(CN)_4]$ b. H_2SO_4 **c. NaCN** d. None

39. . Electrolyte used in chromium plating:

- a. $Cr_2(SO_4)_3$ **b. Cr_2SO_4** c. $Cr(NO_3)_2$ d. $CrSO_4$

40. Chromium plated steel is used to make:

- a. Glass **b. Automobile parts** c. Plastic d. none

ELECTROCHEMICAL SERIES

41. The electrochemical series is a list of elements arranged according to their:

- a) Atomic mass b) Boiling point **c) Standard electrode potentials** d) Number of electrons

42. In the electrochemical series, elements at the top are:

- a) Least reactive metals b) Most reactive non-metals **c) Strong reducing agents** d) Weak oxidizing agents

43. Which of the following is more reactive than copper?

- a) Silver b) Gold **c) Zinc** d) Platinum

44. In the electrochemical series, hydrogen is used as:

- a) A neutral point** b) A strong oxidizing agent
c) A reducing agent d) The least reactive metal





FUEL CELL

45. What is the main product of a hydrogen fuel cell?

- a) Carbon dioxide b) Methane c) Water d) Nitrogen

46. Which of the following is NOT an advantage of fuel cells?

- a) High efficiency b) Low emissions c) Silent operation d) High operating temperature for all types

47. Which of the following is commonly used as a fuel in fuel cells?

- a) Methane b) Hydrogen c) Propane d) Butane

48. What is the role of the electrolyte in a fuel cell?

- a) Burn the fuel b) Transfer electrons c) Conduct ions d) Produce electricity

49. Which of the following is a disadvantage of fuel cells?

- a) No pollution b) Renewable energy source c) High initial cost d) High energy density

50. Fuel cells convert _____ energy into electrical energy.

- a) Thermal b) Kinetic c) Chemical d) Mechanical

51. The by-product of a fuel cell can be used as:

- a) Lubricant b) Fuel for combustion engines

- c) Drinking water in space missions d) Industrial solvent

SECTION B

SHORT QUESTIONS

1. Define electrochemistry.

Ans. The branch of chemistry which deals with the inter conversion of chemical and electrical energy is known as electrochemistry.

Inter conversions:

Chemical energy Spontaneous Redox Reactions Electrical Energy

Non-spontaneous Redox Reactions

Electrical energy chemical Energy

2. What is electrochemical cells also discuss its types?

Ans.

These are the devices in which the inter-conversion of chemical and electrical energy takes place with the help of Redox Reactions.

Types of electrochemical Cells: There are two types of electrochemical cells;

1. Electrolytic cells

2. Galvanic or Voltaic cells

a. Electrolytic cells:

In these cells electrical energy is changed into chemical energy. Non spontaneous redox reactions are carried out in these cell. with the help of electricity. Nelson's cell and Down's cell are example of electrolytic cell.

b. Galvanic or Voltaic cells:

In these cells chemical energy is changed into electrical energy. Spontaneous redox reactions occur in these cells. Electricity is produced in these cells. Daniel's cell and zinc carbon dry cells are examples of galvanic or voltaic cells.

3. Differentiate between Electrolytic and Galvanic cell?

Ans.

Feature	Electrolytic Cell	Galvanic Cell (Voltaic Cell)
Purpose	Converts electrical energy into chemical energy	Converts chemical energy into electrical energy
Type of Reaction	Non -spontaneous (requires external energy)	Spontaneous (produces energy)
Energy Source	Requires external power supply (like a battery)	Generates its own voltage
Electrode Polarity	Anode is positive , Cathode is negative	Anode is negative , Cathode is positive
Electron Flow	From external source to cathode	From anode to cathode through wire
Examples	Electroplating, electrolysis of water	Daniell cell, batteries



4. Briefly discuss the nature of electrochemical process?

Ans. Electrochemical processes involve **chemical reactions that either produce or consume electrical energy**. These processes are at the heart of batteries, electrolysis, corrosion, fuel cells, and more.

5. What is electrolytes and non-electrolytes?

- Electrolyte:**

A substance that conducts electricity when it is dissolved in water or in molten state is called electrolyte. e.g, NaCl, KCl, HCl, NaOH etc.

It is composed of ionic bond.

It includes acid, base and salt.

It can be classified as strong and weak electrolyte

- Non- electrolyte**

A substance that cannot conduct electricity when dissolve in water or in molten state is called non electrolyte. e.g urea, glucose , sucrose , benzene etc.

It is composed of covalent bonds.

It includes carbon containing compounds, fat and sugar.

It can be classified as water soluble and water insoluble compounds

6. Briefly discuss Electrolytic cell?**Ans. Definition:**

An electrochemical cell in which electrical energy is used to drive a chemical reaction is called an electrolytic cell.

Construction:

An electrolytic cell consists of:

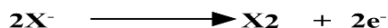
- A vessel containing an electrolyte
- Two inert electrodes
- Battery

Working

On passing electricity the ions move towards their respective electrodes. Cation moves towards cathode and anion moves towards anode. Oxidation occurs at anode and reduction occurs at cathode.

> Chemical reactions:

- At anode:**



- At cathode:**

**7. Define the term electrolysis.**

Ans. Electrolysis is a process in which **electrical energy is used to drive a non-spontaneous chemical reaction**. It is the key process behind things like electroplating, metal extraction, and water splitting. This process involves the movement of ions towards oppositely charged electrodes. Where oxidation occur at anode and reduction occurs at cathode.

8. How lead is produced from electrolysis of molten chloride using platinum or graphite electrodes?

Ans. Electrolysis of molten lead(II) chloride ($PbCl_2$) is an important industrial process used to extract **lead** metal. The process involves breaking down the compound into its constituent elements, **lead** and **chlorine**, using electric current.

- Electrolyte:** Molten lead(II) chloride ($PbCl_2$). The solid is heated until it melts, forming a conductive liquid.

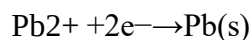
- **Electrodes:** Platinum or graphite electrodes are typically used because they are inert and do not react with the electrolyte or products.

Electrolytic Cell Reaction.

When an electric current is passed through the molten PbCl_2 , the following reactions take place:

At the Cathode (Reduction Reaction):

- The **lead(II) ions (Pb^{2+})** gain electrons (reduction) and are deposited as **solid lead (Pb)**.



At the Anode (Oxidation Reaction):

- The **chloride ions (Cl^-)** lose electrons (oxidation) and are released as **chlorine gas (Cl_2)**.



9. Briefly discuss the electrolysis of dilute aqueous NaCl using platinum electrodes?

Ans. Electrolysis of dilute aqueous NaCl, chloride ions are not oxidized at the anode. Hydroxide ions are oxidized and oxygen gas is produced at the anode. At cathode hydrogen ions get reduced and produce hydrogen gas and at anode hydroxyl ion gets oxidized and forms water and oxygen gas.

Reactions:

At cathode : $4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2$

At anode : $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 4\text{e}^- + \text{O}_2$

10. Briefly discuss the electrolysis of dilute sulphuric acid using platinum electrodes?

Ans. The electrolysis of dilute sulfuric acid (H_2SO_4) using platinum electrodes involves the decomposition of water into hydrogen and oxygen ions when a direct current is passed through the acid.

Electrolyte: Dilute sulfuric acid (H_2SO_4) is used as the electrolyte. The acid dissociates into hydrogen ions (H^+) and sulfate ions (SO_4^{2-}).

Electrodes: Platinum is used as the electrode material because it is inert, meaning it doesn't participate in the reaction but simply provides a surface for the reactions to occur.

At the cathode (reduction): Hydrogen ions (H^+) from the acid migrate to the cathode (negative electrode). Each hydrogen ion gains an electron and is reduced to form hydrogen gas (H_2). $4H^+ + 4e^- \rightarrow 2H_2$

At the anode (oxidation): Hydroxyl ion move towards anode and oxidation take place. The sulfate ions (SO_4^{2-}) do not participate directly in the electrode reactions due to the relatively high stability of the sulfate ion.
 $4OH^- \rightarrow 2H_2O + 4e^- + O_2$

11. Briefly discuss the electrolysis of copper sulphate using platinum electrodes?

Ans. The electrolysis of copper sulfate ($CuSO_4$) using platinum electrodes involves the decomposition of copper sulfate into its constituent ions when an electric current is passed through the solution. Here's a brief overview:

Electrolyte: Copper sulfate solution ($CuSO_4$) dissociates into copper ions (Cu^{2+}) and sulfate ions (SO_4^{2-}) when dissolved in water.

Electrodes: Platinum is used as the electrode material because it is inert, meaning it doesn't react with the electrolyte but provides a surface for the reactions to occur.

At the cathode (reduction): Copper ions (Cu^{2+}) migrate towards the cathode (negative electrode). Each copper ion gains two electrons and is reduced to form copper metal, which gets deposited on the cathode.
 $2Cu^{2+} + 4e^- \rightarrow 2Cu$

At the anode (oxidation): At the anode (positive electrode), water molecules undergo oxidation. Oxygen gas (O_2) is released at the anode, and hydrogen ions (H^+) are produced in the solution.



12. Define electroplating also discuss its purpose?

Ans. It is an electrolytic process used to deposit a thin layer of one metal over another metal.

Purpose of electroplating:

- 1 It is used to increase the beauty of metal for decoration purpose.
- 2 It is used to prevent corrosion.
- 3 It is used for repairing.

13. What are the conditions and procedure of electroplating?

Procedure Of Electroplating

An object can be electroplated by making it cathode in an electrolytic tank containing ions of the plating metal. The plating metal is made anode. On passing electricity through the electrolytic tank, a thin layer of anode metal is deposited on the surface of the object.

Conditions of Electroplating:

Following conditions are required for good deposits during electroplating;

- a. High current density
- b. Low temperature
- c. High conc. Of metal in its electrolyte

14. Briefly discuss Zn plating?

Ans It is the deposition of a thin layer of zinc over another metal by electrolytic process is called zinc plating.

Structure of Electrolytic cell :

The electrolytic cell is used for zinc plating has following components

- a. Container
- b. Electrodes (Zinc metal is used as anode and the object which is to be plated is used as cathode)
- c. Electrolyte (Potassium zinc cyanide $K_2[Zn(CN)_4]$)

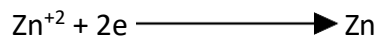
> Mechanism:

- **Ionization equation of electrolyte;** $K_2[Zn(CN)_4] \rightleftharpoons 2K^+ + [Zn(CN)_4]^{2-}$

$[Zn(CN)_4]^{2-} \rightleftharpoons Zn^{+2} + 4CN^{-1}$ **1 React on at anode:**

At anode zinc atoms get oxidized to zinc ions (Zn^{+2}) which are released in solution.

2. React on at cathode: At cathode zinc ions are reduced to zinc atoms which are deposited over the object which is made cathode.

**Prevent on of Hydrolysis**

Sodium cyanide prevents the hydrolysis of electrolyte

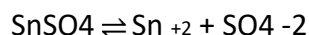
15. Briefly discuss Sn plating?

Ans It is the deposition of a thin layer of tin over another metal by electrolytic process is called zinc plating.

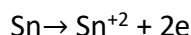
Structure of Electrolytic cell :

The electrolytic cell is used for zinc plating has following components

- b. Container
- b. Electrodes (Tin metal is used as anode and the object which is to be plated is used as cathode)
- c. c. Electrolyte (Stannous Sulphate ($SnSO_4$))

**Mechanism:****Ionization equation of electrolyte;****a. Reaction at anode:**

At anode tin atoms get oxidized to tin ions which are released in solution.

**b. Reaction at cathode:**

At cathode tin ions are reduced to tin atoms which are deposited over the object which is made cathode.

**16. Briefly discuss chromium plating?**

Ans It is the deposition of a thin layer of Cr over another metal by electrolytic process is called zinc plating.

Structure of Electrolytic cell :

The electrolytic cell is used for zinc plating has following components

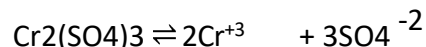
A. Container

d. Electrodes (Cr metal is used as anode and the object which is to be plated is used as cathode)

c. Electrolyte (chromium Sulphate $\text{Cr}_2(\text{SO}_4)_3$)

Mechanism:

- Ionization equation of electrolyte;**

**Reaction at anode:**

At anode Cr atoms get oxidized to Cr ions which are released in solution. $\text{Cr} \rightarrow \text{Cr}^{+3} + 3\text{e}^-$

Reaction at cathode:

At cathode Cr ions are reduced to Cr atoms which are deposited over the object which is made cathode. $\text{Cr}^{+3} + 3\text{e}^- \rightarrow \text{Cr}$

17. Why food cans are tin plated?

Ans. Food cans are tin-plated primarily to prevent corrosion and to protect the contents from contamination. The process involves coating a steel or iron can with a thin layer of tin. Here are the main reasons why tin plating is used for food cans:

1. **1) Corrosion Resistance:** Tin is a corrosion-resistant metal, which helps protect the steel or iron can from rusting. Rust would not only weaken the can but also affect the quality and safety of the food inside.
2. **Non-reactivity with Food:** Tin is relatively non-reactive with most food products, meaning it doesn't interact with the food or alter its taste, color, or nutritional value. This is particularly important for preserving the quality of food over long periods.

18. What is electrochemical series?

Ans. The **electrochemical series** (also known as the **activity series**) is a list of elements (usually metals) arranged in order of their ability to donate electrons (i.e., their relative reactivity). It shows how easily a substance can be oxidized, with the most reactive elements at the top and the least reactive ones at the bottom.

1. **Top of the Series:** The elements at the top are strong reducing agents. They readily lose electrons and undergo oxidation (they have a high tendency to give up electrons). These are usually alkali metals (e.g., lithium, sodium, potassium) and alkaline earth metals (e.g., calcium, magnesium).
 - For example, potassium (K) is a very strong reducing agent, so it is placed at the top.
2. **Bottom of the Series:** Elements at the bottom are weaker reducing agents and tend to gain electrons (they have a high tendency to be reduced). These include noble metals like gold (Au), platinum (Pt), and silver (Ag), which are less reactive and do not easily undergo oxidation.
 - For example, platinum (Pt) is less likely to lose electrons and is placed near the bottom.

19. Define the term battery.

Ans. A **battery** is a combination of one or more **electrochemical cells** connected together to produce a steady and usable **electric current**.

**20. Enlist advantages of fuel cell.**

Ans: High Efficiency

- Fuel cells convert chemical energy directly into electrical energy, often achieving higher efficiencies (40–60%) than traditional combustion engines.

- Clean Energy**

The main by-product of hydrogen fuel cells is **water**, making them environmentally friendly and producing **no harmful emissions** like CO₂ or NO_x.

- Quiet Operation**

Fuel cells operate silently compared to internal combustion engines, making them ideal for use in noise sensitive environments.

Renewable Fuel Sources

Hydrogen, the most common fuel, can be produced from renewable sources like water (via electrolysis) using solar or wind energy.

21. Enlist disadvantages of fuel cell.

Ans: High Cost

Fuel cells are expensive to produce, especially due to the use of precious metals like platinum as catalysts.

- Hydrogen Storage Issues**

Hydrogen, the most common fuel, is difficult to store and transport due to its low density and high flammability.

22. What are the functions of salt bridge in galvanic cell?

- Maintains Electrical Neutrality**

- It allows the flow of ions between the two half-cells to balance the charges that build up due to the redox reactions.



Without it, the reaction would stop quickly due to charge imbalance.

- **Completes the Circuit**

It provides a path for the movement of ions, allowing the electric current to flow continuously through the external wire.

- **Prevents Mixing of Solutions**

The salt bridge allows ion exchange without directly mixing the different solutions, which could otherwise react and stop the cell from working.

SECTION-C

LONG QUESTIONS

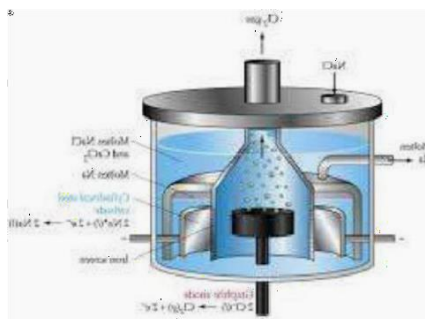
1. Explain electrolysis of molten electrolyte. OR Explain Down's cell ?

Ans. Introduction:

On the large scale sodium metal is produced by the electrolysis of fused sodium chloride. An electrolytic cell called Down's cell is used for this process.

Construction of Down's Cell

The Down's cell is composed of an iron cathode and graphite anode. It also has inverted funnel in which chlorine is collected at the top of cell. There is steel screen between anode and cathode to prevent their contact with each other.



Working:

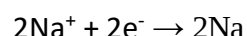
Molten sodium chloride is put into the cell through inlet. Sodium chloride ionizes into its ions through electrolysis. Chloride ions move towards the anode where they lose electrons to form chlorine gas. Chlorine is collected into the funnel at the top of cell. Sodium ions move towards

cathode where they gain electrons to form sodium metal. Sodium metal is collected into the beaker through outlet from cell.

Chemical Reactions:

Following chemical reactions occurs in Down's cell.

Ionization of electrolyte;



2. Explain electrolysis of Brine. OR Explain Nelson cell ?

Introduction:

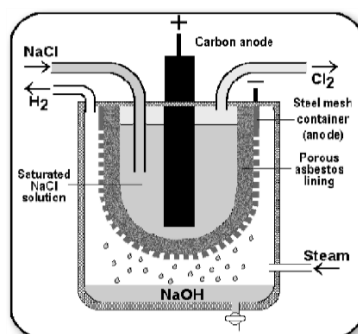
Nelson's cell was discovered by an American chemist Walter Nelson. On industrial scale NaOH is manufactured from brine. Brine is an aqueous solution of salt. For the production of caustic soda we use concentrated aqueous solution of sodium chloride salt.

Construction of Nelson's cell:

Nelson's cell has steel tank which containing graphite anode and perforated iron cathode. It also has asbestos diaphragm layer that prevents connection between anode and cathode. It has beaker which is fitted below iron cathode for the collection of caustic soda

Working:

During electrolysis of brine in Nelson's cell, brine ionizes in its ions. Chloride ions move towards anode where they form chlorine gas by losing electrons. Hydrogen ions from water move towards cathode where they gain electron to form hydrogen gas. These two gases chlorine and hydrogen discharge from cell through outlet. Remaining sodium ions and hydroxyl ions combine with each other to form caustic soda. This is collected in beaker.



Chemical reactions:

- **At anode.** $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- **At cathode.** $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- **Overall reaction.** $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow \text{Cl}_2 + \text{H}_2 + 2\text{NaOH}$

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3. Explain working of Galvanic cell?**Definition:**

Description: It is a type of electrochemical cell in which chemical energy is change into electrical energy with the help of spontaneous redox reaction.

Types:

- There are two types of Galvanic cells:
- 1. Daniel cells 2. Zinc-Carbon dry cell

1. Daniel cells

It is a simplest form of Galvanic cells. It uses spontaneous redox reaction to produce electricity.

Construction:

Daniel cell consist of four main components.

- i. container ii. Electrolyte iii. Electrodes iv. Salt bridge

i. Container: Two separate beakers are used as container.

sulphate as electrolyte. Each beaker is called a half cell.

ii. Electrolytes: one beaker contains aqueous solution of copper sulphate as electrolyte and other beaker contains aqueous solution of zinc sulphate as electrolyte. Each beaker is called a half cell.

iii. Electrodes: Copper plate is used as cathode in copper sulphate solution and zinc plate is used as anode in zinc sulphate solution.

iv. Salt Bridge: It is a inverted “U” shaped glass tube filled with an inert salt like KCl. The ends of the tube are filled with cotton plugs.

THE END

